1 Problem Analysis: CERCETASI

1.1 Problem Statement Overview

The problem asks us to count the number of ways to place n research participants into r research groups, where each group must contain at least st and at most dr participants. The answer is required modulo $M = 10^9 + 7$.

1.2 Key Observations

Observation 1: Normalizing the minimum requirement. If each group must contain at least st participants, we can first assign st participants to each of the r groups. This uses $r \cdot st$ participants. The remaining participants need to be distributed such that each group gets at least 0 additional participants.

Let $n' = n - r \cdot (st - 1) - r = n - r \cdot st$ be the number of participants after normalization. After this transformation:

- Each group must contain at least 0 and at most $dr - st$ additional participants.

Observation 2: Counting distributions without upper bounds. For a fixed number k of non-empty groups (after the transformation, we allow empty groups), the number of ways to distribute n' participants is given by the **Stirling numbers of the second kind**:

$$S(n', k) = \frac{1}{k!} \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} i^{n'}$$

Observation 3: Handling the upper bound via Inclusion-Exclusion. The constraint "at most $dr - st$ per group" means we must exclude distributions where some groups receive more than this limit. Using the principle of inclusion-exclusion:

Let T be the total number of distributions without the upper bound. To count distributions where at most $dr - st$ participants go to each group, we subtract distributions where at least one group exceeds this limit.

For a fixed subset j of groups exceeding the limit, we shift $dr - st + 1$ participants to each group in j and count distributions of the remaining participants. This yields the alternating sum formula implemented in the code.

1.3 Mathematical Formulation

After transformations in the code ($n \leftarrow n - r + 1$, $st \leftarrow st - r + 1$, $dr \leftarrow dr - r + 1$), the answer is:

$$\text{Ans} = \sum_{k=st}^{dr} S(n, k) \cdot \sum_{j=0}^{dr-k} (-1)^j \binom{k+j}{j}$$

The inner sum is precomputed in the array `pref` using the inverse factorials:

$$\text{pref}[m] = \sum_{i=0}^{m} (-1)^i \cdot \frac{1}{i!}$$

The contribution for each k is then:

$$\text{add} = (\text{pref}[dr - k] - \text{pref}[st - k - 1]) \cdot \frac{k^n}{k!} \mod M$$

1.4 Algorithm

1. Precompute factorials, inverse factorials, and modular inverses up to $NMAX$ in $O(NMAX)$ time.
2. Compute the prefix array `pref` where `pref`$[i] = \sum_{j=0}^{i} (-1)^j \cdot \frac{1}{j!}$.
3. For each k from 0 to dr:
   - Compute $k^n \cdot \frac{1}{k!} \mod M$ using fast exponentiation.
   - Retrieve the coefficient sum from the prefix array for the range $[st - k, dr - k]$.
   - Add the product to the answer.
4. Output the final answer.

1.5 Implementation Details

The code implements several key functions:

- `pre()`: Computes factorials, inverse factorials, and modular inverses using the recurrence $\text{inv}[p] = \left(-\left\lfloor \frac{M}{p} \right\rfloor \cdot \text{inv}[M \bmod p]\right) \bmod M$.
- `comb()`: Returns $\binom{n}{k} \mod M$ using precomputed values.
- `pw()`: Fast exponentiation in $O(\log n)$ time.
- `get_stir()`: Computes Stirling numbers of the second kind using the explicit formula (though the main solution uses a more efficient approach).

The main solution loop iterates k from 0 to dr (at most 10^5 iterations), giving an overall complexity of $O(NMAX + r \cdot \log M)$.

1.6 Time and Space Complexity

- **Time Complexity**: $O(NMAX + dr \cdot \log M)$, dominated by precomputation and the final summation loop.
- **Space Complexity**: $O(NMAX)$ for precomputed factorials and prefix array.

1.7 Correctness Argument

The solution is correct because:

1. The transformation $n \leftarrow n - r + 1$ correctly accounts for the minimum requirement by reducing the problem to distributing n labeled balls into any number of groups.
2. Stirling numbers $S(n, k)$ count exactly the distributions of n balls into exactly k non-empty groups.
3. The inclusion-exclusion approach with the prefix sums correctly subtracts distributions exceeding the maximum limit $dr - st$.
4. All modular operations preserve the correct arithmetic modulo $10^9 + 7$.